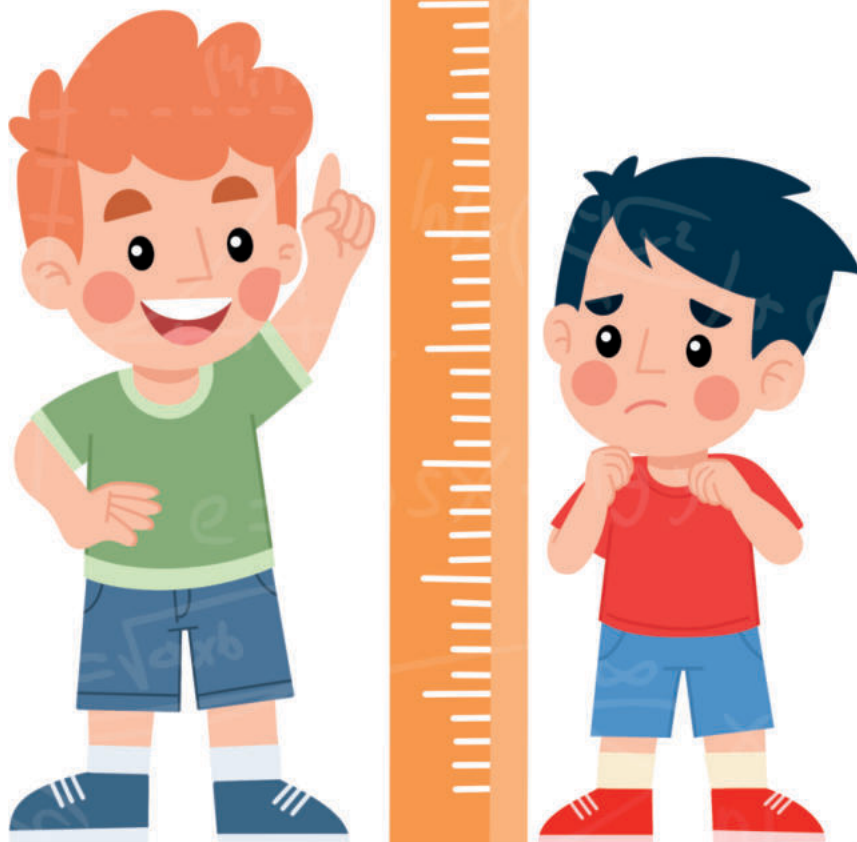


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MEASUREMENT





- **What is length?**

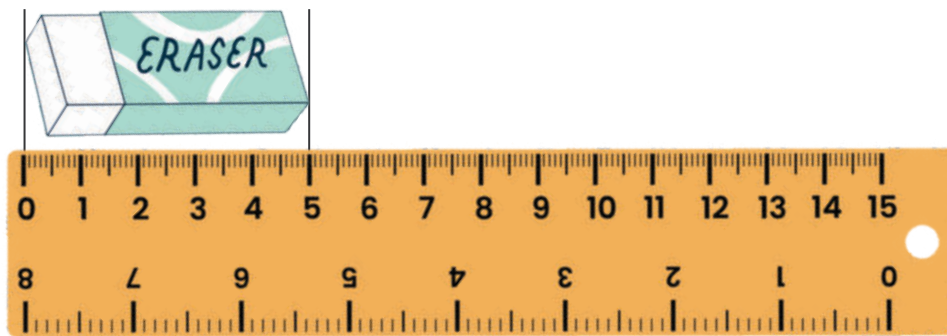
Length is the measure of how long or wide something is.

Let us look at the example below:

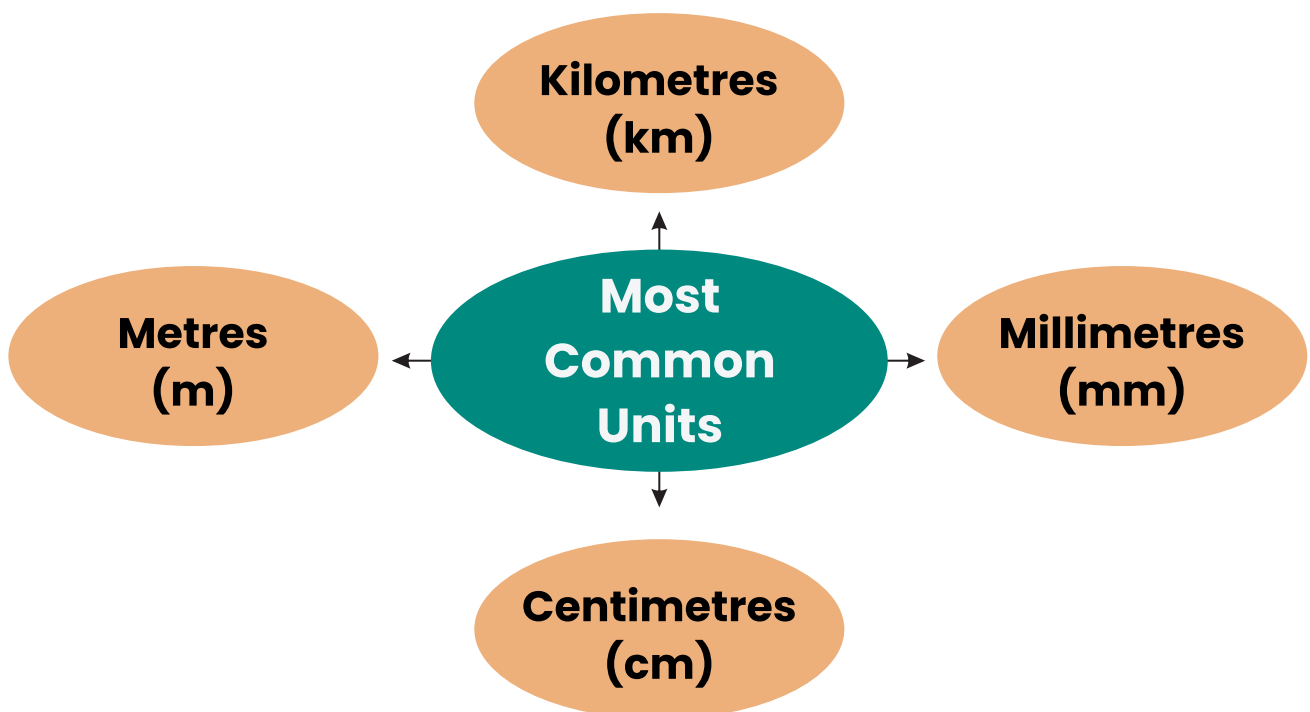
How long is the eraser?

The eraser is **5 cm** from one end to the other.

So the length of the eraser is **5 cm**.



- **Units of Length**



- **Units of Length and their relationship**

Remember:

Kilometres is written as km

Metres is written as m

Centimetres is written as cm

Millimetres is written as mm

The biggest unit of length is kilometres (km).

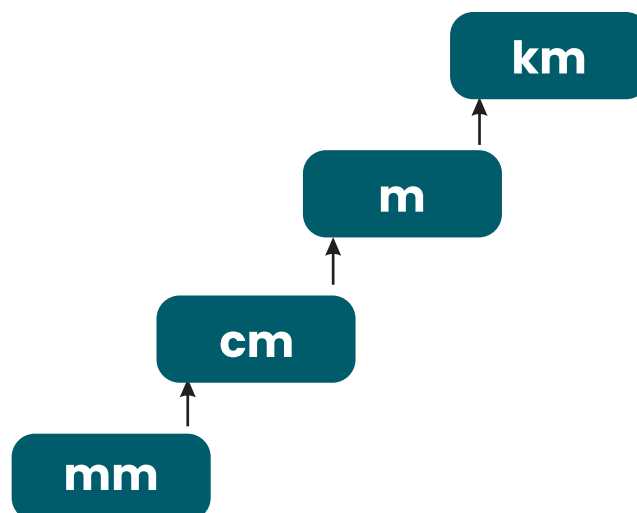
Standard unit of length is metres (m).

Smallest unit of length is millimetres (mm).

1 kilometre = 1000 metres

1 metre = 100 centimetres

- **Units of Length from smaller to larger**



• Conversions

Metres and Centimetres to Centimetres:

To convert 'metres' to 'centimetres,' we multiply the number of metres by 100 and add the number of centimetres.

For Example : Convert 2 m 12 cm into cm

$$1 \text{ m} = 100 \text{ cm}$$

$$2 \text{ m} = 2 \times 100 \text{ cm} = 200 \text{ cm}$$

$$2 \text{ m } 12 \text{ cm} = 200 \text{ cm} + 12 \text{ cm} \\ = 212 \text{ cm}$$

Centimetres to Metres::

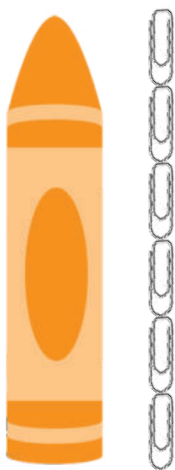
To convert 'centimetres' to 'metres,' we divide the number of centimetres by 100.

Example: Convert 375 cm into metres:

$$1 \text{ m} = 100 \text{ cm}$$

$$375 \text{ cm} \div 100 = 3.75 \text{ m}$$

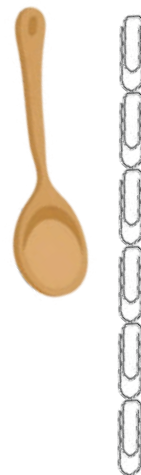
• Measure the lengths of following objects with the help of paperclips.



paper clips



paper clips

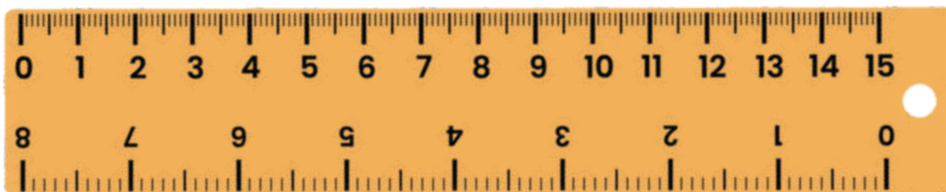
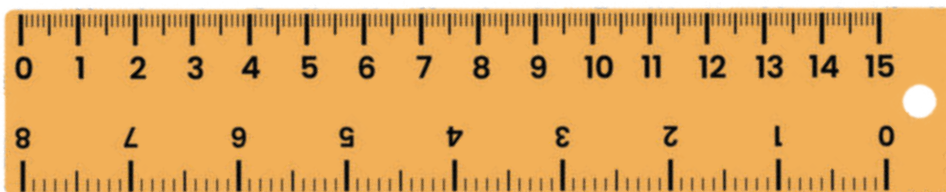


paper clips



paper clips

- **Measure the lengths of the pencils below.**





MEASUREMENT



- **Convert metres into centimetres:**

1. 67 m 20 cm =

2. 17 m =

3. 24 m 30 cm =

4. 5 m =

5. 10 m =

- **Convert centimeters into metres and centimetres.**

1. 3508 cm =

2. 4000 cm =

3. 1235 cm =

4. 768 cm =

5. 930 cm =



• Addition

Case 1:

Lengths to be added are converted into cm.

$$15 \text{ m } 18 \text{ cm} = 1500 \text{ cm} + 18 \text{ cm} = 1518 \text{ cm}$$

$$16 \text{ m } 22 \text{ cm} = 1600 \text{ cm} + 22 \text{ cm} = 1622 \text{ cm}$$

Now, sum

$$\begin{array}{r} \overset{1}{1} \ \overset{1}{5} \ 18 \text{ cm} \\ + 1 \ 6 \ 22 \text{ cm} \\ \hline 3 \ 1 \ 40 \text{ cm} \end{array}$$

Therefore, **15 m 18 cm + 16 m 22 cm = 31 m 40 cm**

Case 2:

$$\begin{array}{r} \text{m} \quad \text{cm} \\ \overset{1}{1} \ \overset{1}{5} \ 18 \\ + 1 \ 6 \ 22 \\ \hline 3 \ 1 \ 40 \end{array}$$

- First cm are added, **18 cm + 22 cm = 40 cm**. This sum is placed under cm column.
- Now metres are added, **15 m + 16 m = 31 m**. This **31 m** is placed under m column.

Hence, sum = **31 m 40 cm**.

• Add the following:

	m	cm
	88	18
	5	60
+	164	80

	m	cm
	18	00
+	23	00

	m	cm
	31	50
+	46	00

	m	cm
	34	02
+	43	06

	m	cm
	68	25
+	36	70

	m	cm
	43	62
+	27	34

	m	cm
	74	50
+	58	70

	m	cm
	47	84
+	6	58

	m	cm
	93	75
+	29	25

	m	cm
	136	45
	49	65
+	17	90

	m	cm
	189	75
	238	85
+	426	68

	m	cm
	87	60
	64	80
+	29	40

• Subtraction

1. Subtract **12 m 36 cm** from **48 m 57 cm**

Case 1:

Minuend and subtrahend are both converted into smaller units.

$$12 \text{ m } 36 \text{ cm} = (12 \times 100) \text{ cm} + 36 \text{ cm} = (1200 + 36) \text{ cm} = 1236 \text{ cm}$$

$$48 \text{ m } 57 \text{ cm} = (48 \times 100) \text{ cm} + 57 \text{ cm} = (4800 + 57) \text{ cm} = 4857 \text{ cm}$$

Now, subtract

$$\begin{array}{r} 48 \text{ m } 57 \text{ cm} \\ - 12 \text{ m } 36 \text{ cm} \\ \hline \end{array} = \begin{array}{r} 4 \text{ } 8 \text{ } 5 \text{ } 7 \text{ cm} \\ - 1 \text{ } 2 \text{ } 3 \text{ } 6 \text{ cm} \\ \hline 3 \text{ } 6 \text{ } 2 \text{ } 1 \text{ cm} \end{array}$$

$$= 3600 \text{ cm} + 21 \text{ cm}$$

$$= 36 \text{ m } 21 \text{ cm}$$

Therefore, **48 m 57 cm - 12 m 36 cm = 36 m 21 cm**



Case 2:

As minuend is greater than subtrahend, the minuend is placed above the subtrahend. Then m and cm are arranged in different columns.

Now, subtract

	m	cm
	48	57
-	12	36
	36	21

- Subtracting cm, **57** cm - **36** cm = **21** cm. It is placed under cm column.
- Subtracting m, **48** m - **12** m = **36** m. It is placed under m column.

Hence, difference = **36** m **21** cm



• Subtract the following:

$$\begin{array}{r} \text{m} \quad \text{cm} \\ 63 \quad 87 \\ - 32 \quad 50 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} \text{m} \quad \text{cm} \\ 36 \quad 75 \\ - 16 \quad 05 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} \text{m} \quad \text{cm} \\ 28 \quad 54 \\ - 12 \quad 30 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} \text{m} \quad \text{cm} \\ 43 \quad 65 \\ - 21 \quad 40 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} \text{m} \quad \text{cm} \\ 90 \quad 85 \\ - 40 \quad 30 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} \text{m} \quad \text{cm} \\ 18 \quad 72 \\ - 14 \quad 20 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} \text{m} \quad \text{cm} \\ 11 \quad 55 \\ - 03 \quad 24 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} \text{m} \quad \text{cm} \\ 12 \quad 75 \\ - 11 \quad 35 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} \text{m} \quad \text{cm} \\ 50 \quad 34 \\ - 37 \quad 58 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} \text{m} \quad \text{cm} \\ 64 \quad 20 \\ - 35 \quad 65 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} \text{m} \quad \text{cm} \\ 73 \quad 38 \\ - 57 \quad 19 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} \text{m} \quad \text{cm} \\ 81 \quad 56 \\ - 43 \quad 15 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} \text{m} \quad \text{cm} \\ 40 \quad 72 \\ - 19 \quad 45 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} \text{m} \quad \text{cm} \\ 80 \quad 72 \\ - 69 \quad 85 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} \text{m} \quad \text{cm} \\ 33 \quad 80 \\ - 09 \quad 69 \\ \hline \\ \hline \end{array}$$



Word Problems

Q1: Sarah has a ribbon that is 12 cm long. She gets another ribbon that is 15 cm long. How long are the ribbons together?



Answer: _____

Q2: A piece of wood is 1 metre long. If 30 cm is cut off from it, how much piece of wood is left?



Answer: _____

Q3: Noor has two pieces of rope. One is 2 metres long and the other is 50 cm long. How long are the ropes if you put them end to end?



Answer: _____

Q4: A runner completes a race track that is 100 metres long. If she stops 25 metres before the finish line. How much has she run?



Answer: _____



MEASUREMENT



Q5: Asma buys two shelves for her room; one is 85 cm long and the other is 95 cm long. What is the total length of both shelves combined?



Answer: _____